

Explainable X-Ray Threat Detection using YOLO11 and Grad-CAM

An Explainable Artificial Intelligence (XAI) based approach for automated detection of prohibited objects in X-ray baggage images.

**Internship : Computer Vision and Deep Learning with Generation AI
(CVDL with GenAI)**

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Project Overview / Introduction

Introduction

The growing volume of passengers travelling through airports, railway stations, and other public places has made baggage security screening a critical component of public safety. Security personnel must inspect thousands of X-ray baggage images very day to identify prohibited objects such as knives, guns, scissors, pliers and other dangerous objects. Manual inspection is time - consuming and prone to human error, especially given the large volume of luggage that needs to be processed.

Recent advancements in deep learning have substantial improved the performance of automated object detection systems. Among the available options, the YOLO (You Only Look Once) family of models is widely used due its ability to detect objects accurately while sustaining real-time processing speed. The latest YOLO11 model delivers high detection accuracy with reduced computational requirements, making it suited for practical security applications.

Despite their strong detection performance, deep learning models often operate as black-box systems, making it difficult to understand why they produce a particular prediction. In security-critical settings, explainability is important because users need assurance that detection are based on meaningful image features rather than irrelevant patterns.

To address the limitations, this project combines Grad-CAM, an Explainable Artificial Intelligence (XAI) technique, with YOLO11. YOLO11 handles the detection of prohibited objects, while Grad-CAM generates visual heatmaps that highlight the regions of the X-ray image responsible for the model prediction. This combination improves both the accuracy and transparency of the detecting system.

Problem Statements and Research Objectives

Problem Statement

Detecting prohibited objects in X-ray baggage images is a vital task in transportation security. Traditionally, security personnel manually inspect X-ray scans to spot dangerous items such as knives, firearms, and other similar threats. However, manual inspection is slow and can be effected by human fatigue, overlapping objects, and the complexity of X-ray imagery, all of which can lead to missed detections or false alarms.

Deep learning-based object detection models provide an efficient alternative by automatically identifying suspicious items with high accuracy. However, these models typically function as black-box systems, making it hard to understand the reasoning behind their outputs. In security-sensitive environments, this lack of explainability can reduce user trust and make it difficult to verify whether the model is focusing on the correct regions of the image.

Therefore, there is a need for a system that not only detects prohibited objects accurately but also provides visual explanations to make the prediction process transparent and trustworthy.

Aim of the Project

The aim of this project is to develop a system that can detect prohibited objects accurately and provide visual explanation

Research Objectives

The main objectives of this project are:

- To develop an automated systems for detecting prohibited objects in X-ray baggage images.
- To train the YOLO11n model for accurate and efficient object detection.
- To evaluate the model using performance metrics such as precision, Recall, Mean Average Precision (mAP).
- To integrate Grad-CAM for generating visual heatmaps that explain the model's predictions.
- To improve the transparency and interpretability of the object detection process.
- To demonstrate how explainable Artificial Intelligence (XAI) can enhance trust I AI-based security systems.

Methodology

Dataset Description

The project uses the X-Ray Baggage Scanner Anomaly Detection dataset sources from Kaggle. The dataset contains annotated X-ray baggage images belonging to 5 prohibited object categories: Gun, Knife, Pliers, Scissors, and Wrench. These labelled images were used to train, validate and evaluate the YOLO11n object detection model. The trained model is capable of detecting all five object categories, and Grad-CAM is applied to provide visual explanations of the model predictions.

Dataset Name:

X-Ray Baggage Scanner Anomaly Detection

Source:

Kaggle

Dataset Link:

<https://www.kaggle.com/datasets/orvile/x-ray-baggage-anomaly-detection>

Classes:

- Gun
- Knife
- Pliers
- Scissors
- Wrench

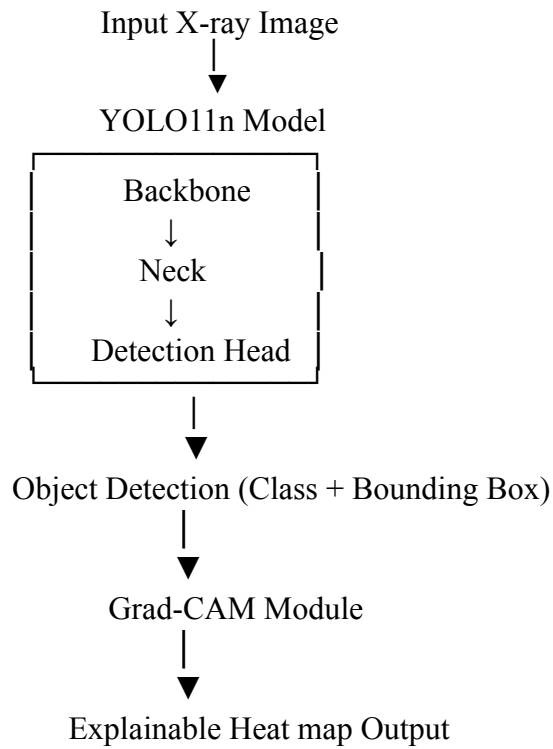
Model Architecture

The proposed model uses YOLO11n for detecting prohibited objects in X-Ray baggage images and Grad-Cam is used to explain the model's predictions. YOLO11n was selected because its a lightweight model which provide a good detection accuracy and it require less computational power, which make it suitable for real-time applications.

The architecture of YOLO11n consist of 3 main parts: Backbone, Neck, Detection Head. The Backbone extracts important features from the input X-ray images. The Neck combine the feature maps from different layers, which helps in detecting object of different sizes. The Detection Head then predicts the object class, bounding box coordinates, and confidence score for each detected object.

Once the object are detected, Grad-CAM is applied to generate a heat map showing the regions of the image that influenced the models prediction. This visualisation helps to clarify why the model detected a particular object and makes the detection process more transparent and reliable.

Proposed Model Architecture



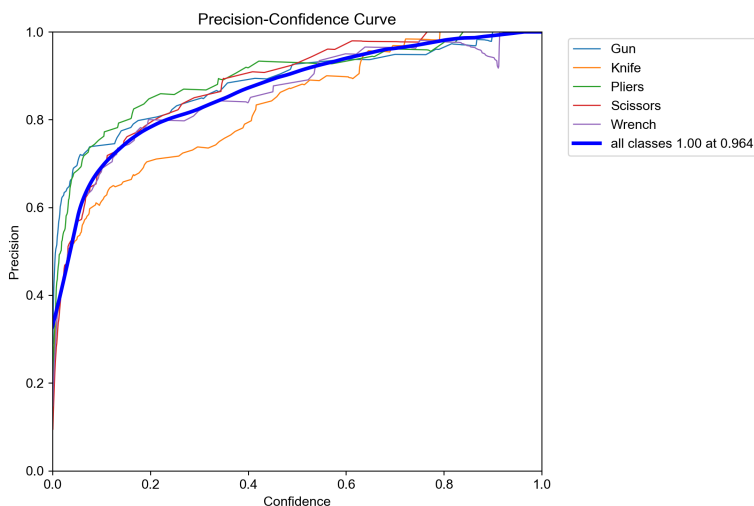
Evaluation Metrics

The performance of the YOLO11n model was evaluated using standard object detection metrics. These metrics measure the accuracy and reliability of the model in detecting objects from X-ray baggage images

Precision

Precision measures the percentage of correctly detected objects out of all the objects predicted by the model. A higher precision value indicated fewer false positive detections.

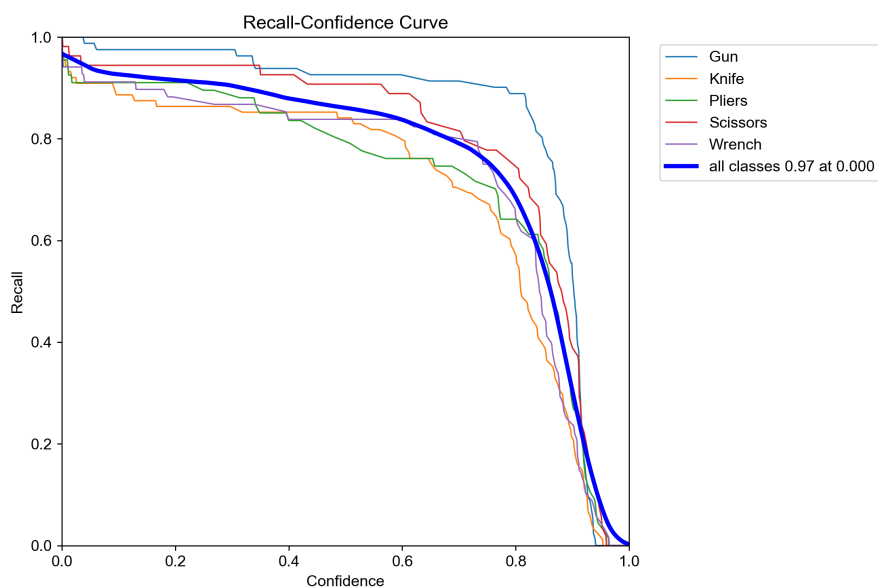
$$\text{PRECISION} = \text{TP} / \text{TP} + \text{FP}$$



Recall

Recall measures the percentage of actual objects that are correctly detected. A higher value indicates that some objects are missed during detection

$$\text{RECALL} = \text{TP} / \text{TP} + \text{FN}$$



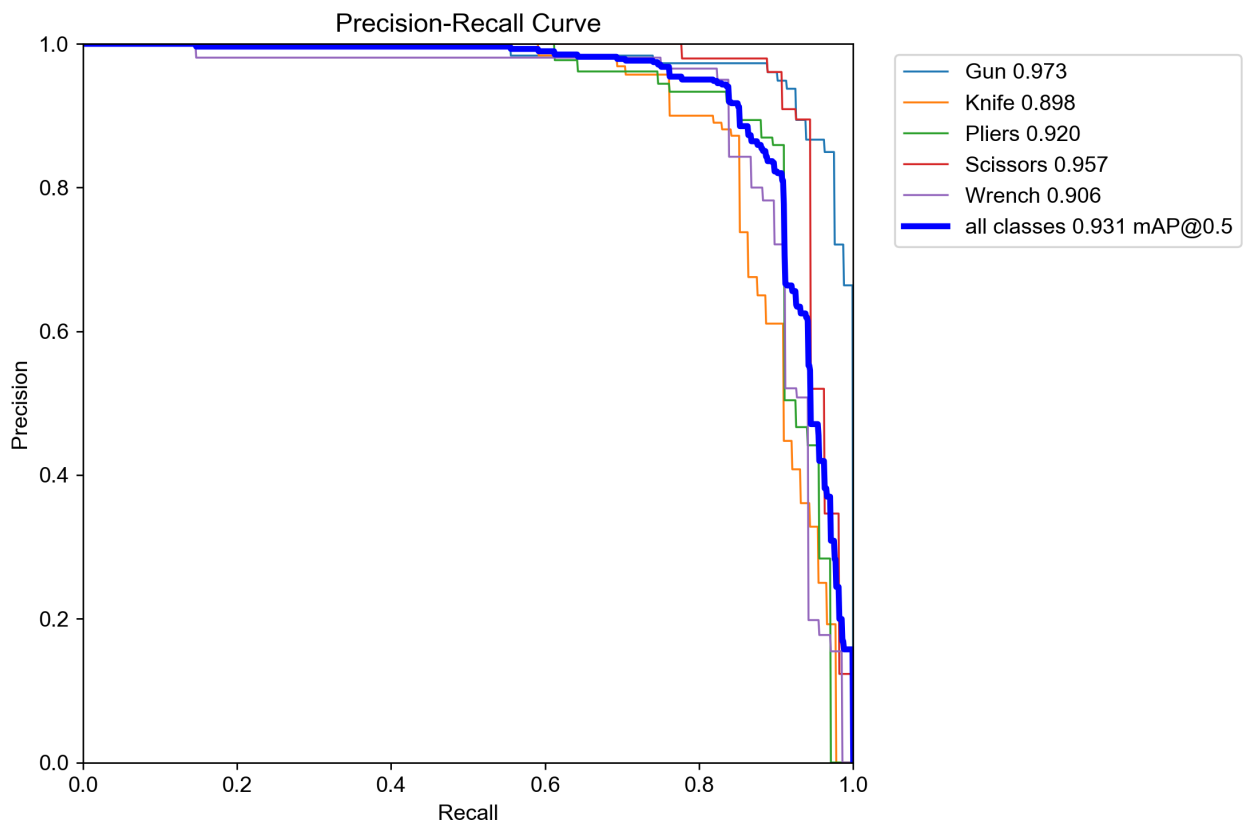
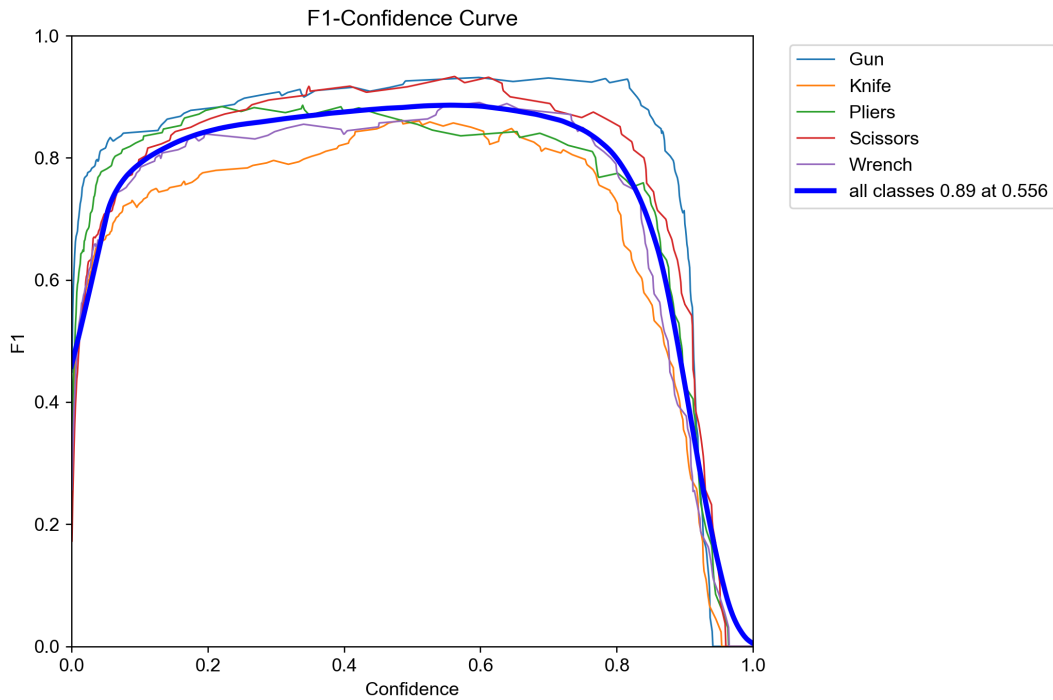
Mean Average Precision(mAP)

Mean Average Precision (mAP) is one the most important metrics used to evaluate detection models. It summarises the detection performance by considering both precision and recall across all object classes. A higher mAP value indicates better overall detection accuracy

F1-Score

The F1-score is the harmonic mean of precision and recall. It provides a balanced measure of the model performance by considering both false positives and negatives.

$$\text{F1 SCORE} = \frac{2 * \text{PRECISION} * \text{RECALL}}{\text{PRECISION} + \text{RECALL}}$$



Experimental Setup

The experiments are conducted to evaluate the performance of the proposed YOLO11n-based explainable X-ray thread detection system. The model was trained using the X-Ray baggage Scanner Anomaly Detection dataset containing five object classes: Gun, Knife, Pliers, Scissors, and Wrench. The dataset was divided into training, validation and testing sets.

The implementation was carried out using Python with the Ultralytics YOLO11n framework. Training and evaluation were performed using PyTorch, while OpenCV was used for Image Processing. Grad-CAM was integrated to generate heat maps for explaining the model predictions.

The model was trained for 50 epochs with an image size of 640 * 640 pixels and a batch of size 16. During training, the model performance was evaluated using Precision, Recall, mAP@50 and mAP@50-95. After training, the best-performing model was used to perform object detection and generate Grad-CAM visualizations.

Experimental Configuration

Parameter	Value
Model	YOLO11n
Framework	Ultralytics YOLO11
Programming Language	Python
Deep Learning Library	PyTorch
Image Processing	OpenCV
Explainability Method	Grad-CAM
Image Size	640 * 640
Batch Size	16
Epochs	50
Dataset	X-Ray Baggage Scanner Anomaly Detection
Number of Classes	5

Results

The proposed YOLO11n - based threat detection model was successfully trained and evaluated on the X-Ray Baggage scanner Anomaly Detection dataset. The model demonstrate good detection performance for all five prohibited object classes. The experimental results were analysed using standard object detection metrics, confusion matrix, prediction outputs and Grad-CAM visualizations.

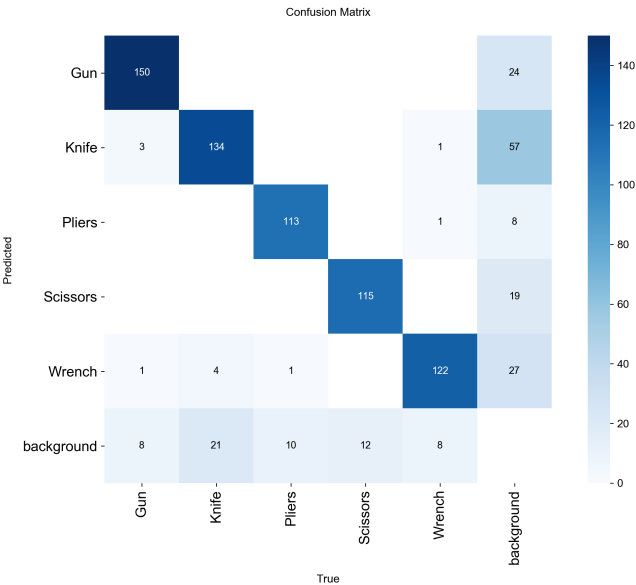
Performance Metrics

The performance of the model was evaluated using Precision, Recall, mAP@50 and mAP@50-95. The results indicated that the trained model can accurately detect prohibited objects while maintaining a good balance between precision and recall.

METRIC	APPROXIMATE VALUE
Precision	0.92
Recall	0.86
mAP@50	0.93
mAP@50-95	0.73

Confusion Metrics Analysis

The confusion matrix was used to evaluate the classification performance of trained model. It show how accurately the model classified each object category. Most samples were accurately classified but only few misclassifications between visually similar objects. This indicated that the model learned meaningful feature from X - ray images and achieved reliable detection performance.

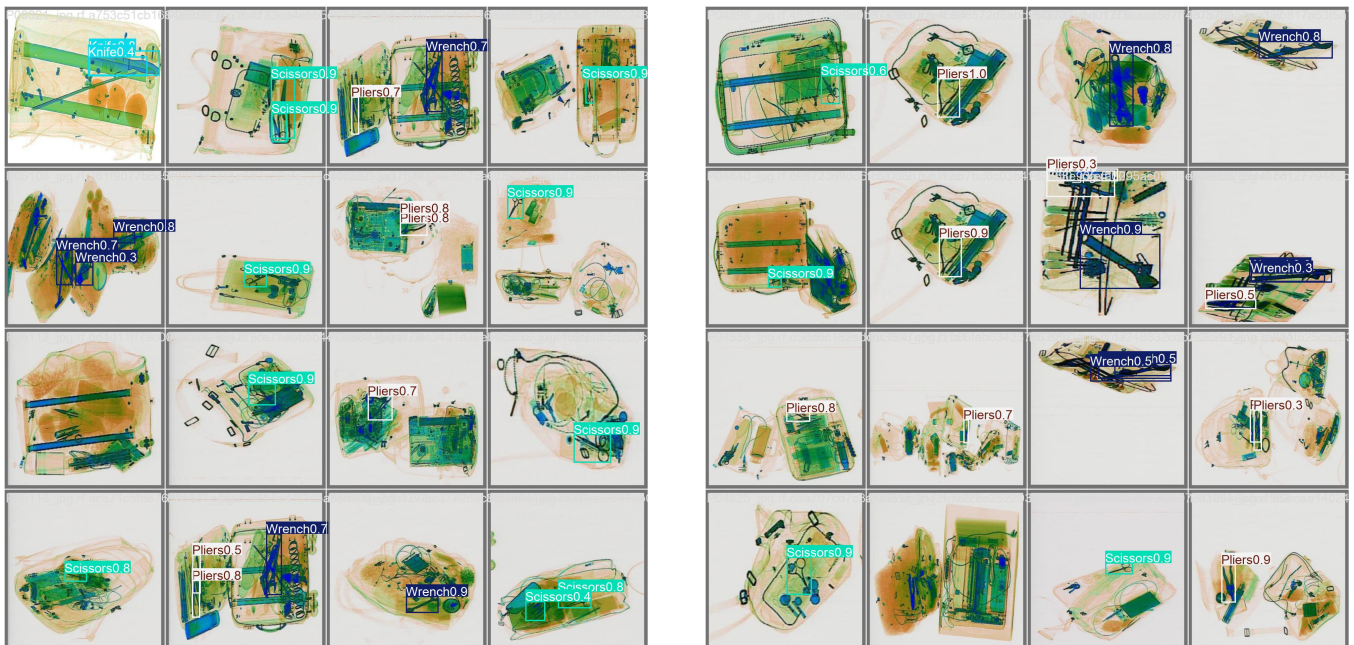


Explainability using Grad-CAM

To improve the interpretability of the detection model, Grad-CAM was applied to visualise the regions responsible for the model predictions. The generated heatmaps highlight the important areas of the X-ray image that contributed to the detection of prohibited objects. These visualizations shows that the model focuses primarily on the actual threat object rather than irrelevant background regions, increasing confidence in the model's predictions

Detection Results

The trained model was tested on unseen X-ray baggage images. The prediction results show that the model successfully detected prohibited objects by generating accurate bounding boxes, class labels, and confidence scores. These results demonstrate the effectiveness of the proposed system in identifying multiple threat objects in different baggage images



Comparative Analysis

The proposed YOLO11n model was selected because it provides a good balance between detection accuracy and computational efficiency. Compared with larger YOLO variants, YOLO11n requires fewer parameters and faster inference while maintaining competitive detection performance. The integration of Grad-CAM further improves the model by providing visual explanations for its predictions, making the system more transparent and reliable for X-ray baggage threat detection.

Conclusion

The project presented an explainable X-ray baggage threat detection system using YOLO11n and Grad-CAM. YOLO11n successfully detected prohibited objects with high accuracy, while Grad-CAM provided visual explanations highlighting the regions responsible for each prediction. The integration of object detection and explainable AI improves the transparency and reliability of automated security screening systems. In future work, the proposed system can be extended to detect additional threat categories and deployed for real-time airport baggage screening.